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Superseding
MIL–I–7069
29 December 1950

MILITARY SPECIFICATION

INDICATORS, TACHOMETER, ELECTRIC, PERCENT SPEED TYPE, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the general requirements for an electric tachometer indicator for use on aircraft with turbine engines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

DD–G–451—Glass, Flat and Corrugated, for Glazing, Mirrors, and Other Uses

MILITARY

MIL–P–116—Preservation, Methods of
MIL–C–5015—Connectors, Electric, "AN" Type
MIL–E–5272—Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL–C–5541—Chemical Films for Aluminum and Aluminum Alloys
MIL–L–6085—Lubricating Oil, Aircraft Instrument, Low Volatility
MIL–S–6872—Soldering Process, General Specification for
MIL–S–7742—Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL–P–7936—Parts and Equipment, Aeronautical, Preparation for Delivery
MIL–A–8625—Anodic Coatings, for Aluminum and Aluminum Alloys
MIL–G–9398—Generator, Tachometer, High Temperature, Two-Pole, Aircraft, Type MB–6
MIL–L–25467—Lighting, Integral, Instrument, General Specification for
MIL–C–26482—Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL–G–26611—Generator, Tachometer, GEU–7/A, Miniature
MIL–L–27160—Lighting, White, Integral, Instrument, General Specification for
MIL–D–70327—Drawings, Engineering and Associated Lists

STANDARDS

FEDERAL

Federal Standard No. 1—Standard for Laboratory Atmospheric Conditions for Testing
Federal Standard No. 595—Colors

MILITARY

MIL–STD–130—Identification Marking of U.S. Military Property
MIL–STD–143—Specifications and Standards, Order of Precedence for the Selection of
MIL–STD–704—Electric Power, Aircraft Characteristics and Utilization of
MS28105—Window, Dial—Aircraft Instrument Cover, Glass
MS33558—Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586—Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification. The indicators furnished under the detail specification which forms a part of this specification shall be a product which has been tested and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Selection of materials. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Standard MIL–STD–143, except as provided in 3.2.1 and 3.2.2.

3.2.1 *Commercial parts.* Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.
(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 *Standard parts.* With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 *Critical materials.* Noncritical materials shall be used where practicable. Where the use of a critical material is essential to conform to specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 *Metals.* Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.2.1 *Dissimilar metals.* Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in Standard MS33586 shall not be used in intimate contact with each other.

3.3.3 *Nonmagnetic materials.* Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.4 *Fungus-proof materials.* Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 *Corrosive fumes.* The materials used in the indicator shall not liberate deleterious fumes.

3.4 Design and construction. The indicator shall conform to the detail specification and shall consist of a suitable "magnetic drag" type indicating mechanism driven by a self-starting, three-phase synchronous motor. Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable. The indicator shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service. The indicator shall be designed for operation with generators conforming to Specification MIL–G–26611 or MIL–G–9398. It shall be possible to operate two indicators from one generator without affecting the operation of either indicator.

3.4.1 *Stops.* The indicator mechanism shall be provided with a stop, which, upon reverse operation of the indicator motor, will stop the pointer motion after traveling through an arc equal to not less than 2 percent nor more than 4 percent.

3.4.2 *Maintenance.* The design of the indicator shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance using those tools and items of maintenance equipment which are normally available as commercial standards.

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3.5 Performance. The indicators shall conform to the following requirements when subjected to the applicable tests of section 4.

3.5.1 *Starting at room temperature.* When tested in accordance with 4.7.2, the indicated speed shall not exceed 180 revolutions per minute (rpm) when the indicator starts to run at synchronous speed.

3.5.2 *Scale error.* When tested in accordance with 4.7.3, the scale errors shall not exceed the specified tolerances. At constant speed, the pointers shall not oscillate over a range greater than 0.5 percent up to 20 percent r.p.m. nor greater than 0.3 percent from 20 to 110 percent r.p.m.

3.5.3 *Friction.* With the indicator set on the 5-, 20-, 40-, 70-, 85-, and 100-percent graduations, as specified in 4.7.4, the change indication shall not exceed 1.5 percent at the 5-percent graduation and 0.8 percent at the 20-percent graduation. The change in indication at the other graduations shall not exceed 0.3 percent for a concentric pointer indicator nor exceed 0.5 percent for indicators employing a subdial.

3.5.3.1 *Indicator position error.* While operated at 100 percent with the indicator in the normal position, the maximum deviation of the readings from the original reading shall not exceed 0.3 percent while held in the four positions specified in 4.7.4.1.

3.5.4 *Pointer alignment.* Indicators with a subdial shall be operated to indicate 90 and 96 percent r.p.m. of the main pointer. The subdial shall indicate 0 and 6, respectively, within 0.20 percent (see 4.7.5).

3.5.5 *Fogging.* When tested as specified in 4.7.6, the indicator shall not show evidence of water or oil fog inside the cover glass.

3.5.6 *Immersion (leakage.)* When tested as specified in 4.7.7, no evidence of leakage shall be found.

3.5.7 *Indicator starting at low temperature.* When tested in accordance with 4.7.8, the indicated speed shall not exceed 180 r.p.m. when the indicator starts to run at synchronous speed.

3.5.8 *Scale error at −55° C.* While operated at −55° C. as specified in 4.7.9, the change in reading of the indicator from the room temperature readings at any point shall not exceed the specified values.

3.5.9 *Scale error at −35° C.* While operated at −35° C. as specified in 4.7.10, the change in reading of the indicator from the room temperature readings at any point shall not exceed the specified values.

3.5.10 *Indicator scale error at high temperature.* While operated at +70° ±2° C., the change in reading from the room temperature reading at any test point shall not exceed the values in 4.7.11.

3.5.11 *Dielectric strength.* The indicator shall be tested at 500 volts as specified in 4.7.12. No breakdown of insulation or permanent damage shall result.

3.5.12 *Illumination.* The illumination of the indicator shall conform to the detail specification (see 4.7.13).

3.5.13 *Magnetic effect.* When tested as specified in 4.7.14, the deflection of the compass shall not exceed 2 degrees.

3.5.14 *Vibration error.* When tested as specified in 4.7.15, the readings with vibration shall not vary more than 0.8 percent from the readings without vibration. The amplitude of the pointer oscillation shall not exceed 0.8 percent. The reading after vibration shall not differ from the reading before vibration by more than 0.3 percent.

3.5.15 *Indicator current.* When tested as specified in 4.7.16, no one of the three-line currents drawn shall exceed 0.30 ampere.

3.5.16 *Indicator temperature rise.* While operated at 21±0.1 volts 4,200±5 c.p.m. for a period of 1 hour, the temperature rise shall not exceed 60° C. as specified in 4.7.17.

3.5.17 *Low temperature exposure.* When tested as specified in 4.7.18, there shall be no damage which would adversely affect subsequent operation.

3.5.18 *High temperature exposure.* After exposure at 71° C. for 48 hours, the indicator shall pass the tests specified in 4.4.1 (see 4.7.19).

3.5.19 *Temperature shock.* After exposure at 85° C. for 4 hours then at −40 C. for 4 hours, the indicator shall not leak as a result of the exposure (see 4.7.20).

3.5.20 *Vibration failure.* When tested as specified in 4.7.21, the reading for increasing speeds only shall not differ from the reading before vibration by more than 0.5 percent. No

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looseness in the mechanism or damage to the indicator shall result from the vibration.

3.5.21 *Indicator overspeed and reverse operation.* After being operated at a speed of 5,000 r.p.m. for 5 minutes, then in the reverse direction of the generator at 2,500 r.p.m. for 1 minute, the indicator readings shall be determined at 20 and 90 percent r.p.m. The change in indications before and after the test shall be not more than 0.5 percent (see 4.7.22).

3.5.22 *Salt spray.* After exposure to salt spray for 168 hours, the indicator shall pass the immersion and dielectric tests (see 4.7.23).

3.5.23 *Life.* After operation for 300 hours as specified in 4.7.24, the change in readings at 20 and 90 percent r.p.m. before and after this test shall be not more than 0.8 percent. No looseness or damage shall occur as a result of the 300-hour operation.

3.6 Case. The outline dimensions of the indicator case shall conform to the requirements of the detail specification. The case shall be hermetically sealed and the design shall be such that the mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures. The hermetic seal shall be a perfectly closed and airtight seal made between metallic and metallic, or metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

3.6.1 *Cover glass.* The cover glass shall conform to Specification DD–G–451 and be of the size specified in Standard MS28105.

3.6.1.1 *Dial.* The dial shall be as specified in the detail specification. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be located so that they will not interfere with any dial marking or prevent locating any dial marking in the proper location.

3.6.1.2 The distance between the inside surface of the cover glass and the indicator dial shall be the minimum practicable. On indicators with concentric pointers, the distance shall not exceed 0.188 inch. On single pointer indicators, with or without a subdial, this distance shall not exceed 0.125 inch.

3.6.2 *Filling medium.* The filling medium shall be of at least 98 percent purity, free of dust particles, and containing not more than 0.006 milligram of water vapor per liter (dewpoint −65° C.) at the filling pressure. The filling medium shall be a mixture of 90 percent nitrogen and 10 percent helium or a filling medium approved by the procuring activity. The absolute pressure of the filling medium in the case shall be approximately one atmosphere.

3.7 Screw threads. Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL–S–7742.

3.8 Electrical power requirements. The indicator lighting shall operate from an alternating current (a.c.) power supply in accordance with Standard MIL–STD–704.

3.8.1 *Wiring.* The three leads from the indicator windings shall be designated as B, A, and C (conforming to phases 1, 2, and 3). The leads shall be connected to the B, A, and C pins of the electrical receptacle in such a manner that when three-phase power with phase rotation sequence of B–A–C is applied, the resultant indicator reading is positive. Electrical connections between indicators and generators shall conform to figure 1.

3.8.2 *Electric connector.* The electric connector used in the construction of the indicator shall be in accordance with Specification MIL–C–5015 or MIL–C–26482 as specified in the detail specification.

3.9 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL–D–70327.

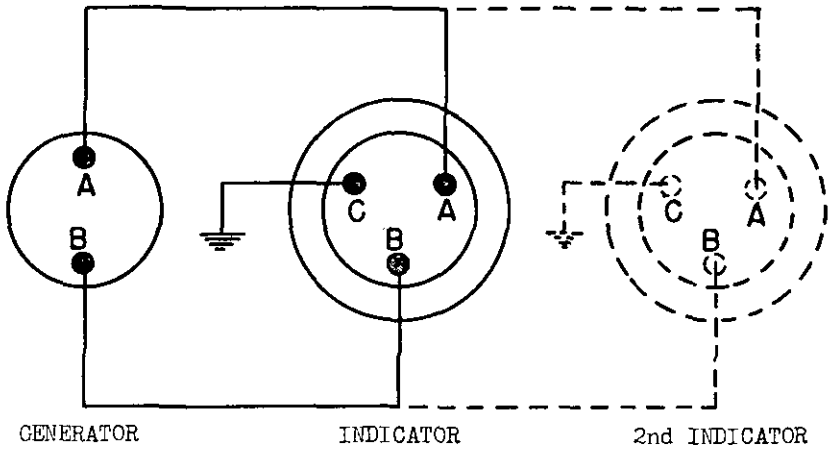
3.10 Physical requirements.

3.10.1 *Weight.* The weight of the indicator shall be as specified in the detail specification.

3.10.2 *Finish.* Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.10.2.1 *Aluminum-alloy parts.* Where practicable, aluminum-alloy parts shall be covered with an anodic film conforming to Specifica-

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THE WIRING DIAGRAM SHOWS THE METHOD OF CONNECTION WHEN THE GENERATOR DRIVE SHAFT TURNS COUNTERCLOCKWISE (WHEN VIEWED FROM THE SHAFT END). IF THE ROTATION IS CLOCKWISE, REVERSE TWO LEADS AT THE GENERATOR PLUG.

IF TWO INDICATORS ARE USED, THEY SHOULD BE CONNECTED IN PARALLEL WITH THE RESPECTIVE GENERATOR AS SHOWN.

FIGURE 1. *Wiring diagram.*

tion MIL–A–8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL–C–5541.

3.10.3 *Soldering.* Soldering shall be performed in accordance with Specification MIL–S–6872.

3.10.4 *Lubrication.* The lubricating oil used shall be in accordance with Specification MIL–L–6085 and shall be sufficient for at least 300 hours of operation.

3.11 Marking. All markings shall be durable to withstand usage encountered in service. The form of the letters and numerals shall conform to Standard MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.11.1 *Dial.* Unless otherwise specified in the detail specification, all visible portions of the dial shall be lusterless black, Color No. 37038 of Federal Standard No. 595 and all markings on the dial shall be lusterless white, Color No. 37875 of Federal Standard No. 595.

3.11.1.1 *Visibility of dial.* All markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.11.1.2 *Illumination of dial.* The indicator shall be lighted, if required, in accordance with Specification MIL–L–25467 or MIL–L–27160, as specified in the detail specification.

3.11.2 *Identification of product.* A suitable nameplate shall be securely attached to the indicator case, and shall be marked in accordance with Standard MIL–STD–130.

3.12 Installation instructions. The installation instruction, when applicable, shall be as specified in the detail specification.

3.13 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all

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inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the indicator shall be classified as follows:

(a) Qualification tests (see 4.3).

(b) Acceptance tests (see 4.4).

4.3 Qualification tests. The qualification tests of the indicator shall consist of all tests of this specification specified under 4.7.

4.3.1 *Qualification test samples.* The qualification test samples shall consist of three indicators of each manufacturer's part number submitted as samples for qualification testing. After obtaining authorization for submittal from the activity responsible for qualification, the samples shall be forwarded to the testing laboratory designated in letter of authorization.

4.4 Acceptance tests. The acceptance tests of the indicator shall consist of the individual tests and the sampling plans and tests (see 4.4.1 and 4.4.2).

4.4.1 *Individual tests.* The individual tests of the indicator shall consist of the following tests conducted on each indicator:

Examination of product	(4.7.1)
Starting test at room temperature	(4.7.2)
Scale error	(4.7.3)
Friction	(4.7.4)
Indicator position error	(4.7.4.1)
Pointer alinement	(4.7.5)
Immersion (leakage)	(4.7.7)

4.4.2 *Sampling plans and tests.*

4.4.2.1 *Sampling plan A.* Samples shall consist of 3 indicators selected at random from each lot of 100 or less indicators which have passed the individual tests. A lot shall consist of identical indicators with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Indicators which have been subjected to the sampling plan A tests shall not be delivered on contract until they have been rebuilt and resubmitted to all individual tests.

4.4.2.1.1 *Sampling plan A tests.* The sampling plan A tests of the indicators shall consist of the following tests conducted on each sample:

Indicator starting at low temperature	(4.7.8)
Scale error at −55° C	(4.7.9)
Scale error at −35° C	(4.7.10)
Indicator scale error at high temperature	(4.7.11)
Dielectric strength	(4.7.12)
Illumination	(4.7.13)
Magnetic effect	(4.7.14)
Vibration error	(4.7.15)

4.4.2.2 *Sampling plan B.* Unless otherwise specified (see 6.2), 3 indicators selected at random from the first 15 indicators produced on contract and submitted within 10 days after manufacture shall be forwarded to the laboratory specified (see 6.3). Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Submitted by (name) (date) for test in accordance with Contract No.

4.4.2.2.1 *Sampling plan B tests.* The sampling plan B tests of the indicators shall consist of the following tests conducted on each sample:

Sampling plan A	(4.4.2.1.1)
Indicator current	(4.7.16)
Indicator temperature rise	(4.7.17)
Low temperature exposure	(4.7.18)
High temperature exposure	(4.7.19)
Temperature shock	(4.7.20)
Vibration failure	(4.7.21)
Indicator overspeed and reverse operation	(4.7.22)
Salt spray	(4.7.23)
Life	(4.7.24)

4.5 Rejection and retest. The instructions on reworking or resubmitting rejected items shall be as specified in the general provisions of the contract.

4.6 Test conditions.

4.6.1 *Atmospheric conditions.* Unless otherwise specified, all tests required by this speci-

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fication shall be made in accordance with the requirements of Federal Standard No. 1.

4.6.2 *Attitude.* Unless otherwise specified, the indicator shall be tested in its level position in which the upper mounting holes are on a horizontal line and the face is vertical.

4.6.3 *Electrical power.* Unless otherwise specified, voltages noted herein are alternating current root-mean-square values. The voltmeter used to measure voltages shall be of a type which is free from frequency errors.

4.6.4 *Tapping.* Unless otherwise specified, the indicator shall be vibrated at 3600 cycles per minute (c.p.m.) with maximum amplitude of 0.002 inch before a test reading is taken.

4.6.5 *Vibration stand.* Whenever a vibration stand is specified, it shall be a device which will vibrate at any desired frequency between 5 and 50 cycles per second (c.p.s.) and shall subject the indicator to such vibration that a point on the vibration stand shall describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.7 Test methods.

4.7.1 *Examination of product.* Each indicator shall be examined externally to determine conformance to the applicable drawings and to all the requirements of this specification not covered by tests.

4.7.2 *Starting test at room temperature.* The indicator shall be connected to a variable frequency power supply. The voltage of the supply in volts shall be equal to 0.0035 times the frequency in cycles per minute. The frequency of the supply shall be increased from 0 cycles per minute, and the speed at which the indicator starts to run at synchronous speed shall be noted. This speed shall not exceed 180 r.p.m. indicated speed.

4.7.3 *Scale error.* Two single indicators shall be connected, as shown on figure 1, to a generator conforming to Specification MIL–G–26611, and tested at points of the scale indicated in table I. The test shall be made by driving the generator at the speed specified to produce these readings, first with speeds increasing then with speeds decreasing. With speeds increasing, the speed shall be brought up to, but not exceeding the speed specified to give the desired reading, and with speeds decreasing, the speed shall be brought down to, but shall not fall below the speed specified to give the desired reading. At the option of the contractor, table II may be substituted for table I whenever it is specified.

4.7.3.1 The errors at the test points shall not exceed the tolerances specified in the applicable table. When the speed is held constant at any point of the scale, the pointer shall not oscillate over a range greater than 0.5 percent up to 20 percent r.p.m. nor greater than 0.3 percent from 20 to 110 percent r.p.m.

4.7.4 *Friction.* The indicator shall be set on the 5-, 20-, 40-, 70-, 85-, and 100-percent graduations, respectively. After each setting, the indicator shall be tapped and the change in indication noted. This change shall not exceed 1.5 percent at the 5-percent graduation and 0.8 percent at the 20-percent graduation. The change in indication at the other graduations shall not exceed 0.3 percent for a concentric pointer indicator and shall not exceed 0.5 percent for indicators employing a subdial. During the friction test, the indicator shall be in the normal upright position (dial vertical) and shall not be tapped until the pointer has come to rest.

4.7.4.1 *Indicator position error.* The indicator shall be operated at 100 percent with the indicator in the normal (dial vertical) position. The indicator reading shall be noted. The indicator shall then be held in the following positions:

(a) Rotated from normal position through 15 degrees to the rear (dial titled 15 degrees back from vertical position).
(b) Rotated from normal position through 15 degrees to the front (dial tilted 15 degrees forward from vertical position).
(c) Rotated from normal position through 15 degrees to the right (dial vertical).
(d) Rotated from normal position through 15 degrees to the left (dial vertical).

The maximum deviation of the four latter readings from the original reading shall not exceed 0.3 percent.

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TABLE I. *Calibration scale errors*

Generator drive shaft speed	Correct indicated speed	Scale error tolerance	Change in scale error at specified temperature (max.)			
		Room temperature	−35° C.	−55° C.	+70° C.	0° C.
R.p.m.	Percent r.p.m.	±percent r.p.m.	±percent r.p.m.			
0*	0	0.5	0.5	0.5	0.5	0.4
200*	4.76	.5	.5	.5	.5	.4
400*	9.52	.5	.5	.5	.5	.4
600	14.29	.5				
800*	19.05	.5	.5	.5	.5	.4
1,200	28.57	.6				
1,600*	38.10	.8	.6	.7	.7	.4
2,000	47.62	.8				
2,400*	57.14	.8	.6	1.0	1.0	.5
2,800	66.67	.8				
3,200*	76.19	.5	.7	1.3	1.3	.5
3,400*	80.95	.5	.8	1.4	1.4	.5
3,600*	85.71	.5	.8	1.5	1.5	.5
3,800*	90.48	.5	.9	1.5	1.5	.5
4,000*	95.24	.5	.9	1.6	1.6	.5
4,200*	100.00	.5	.9	1.6	1.6	.5
4,400*	104.76	.5	1.0	1.8	1.8	.6
4,600	109.52	.5				

*See 4.7.9.

TABLE II. *Calibration scale errors*

Generator drive shaft speed	Correct indicated speed	Scale error tolerance	Change in scale error at specified temperature (max.)			
		Room temperature	−35° C.	−55° C.	+70° C.	0° C.
R.p.m.	Percent r.p.m.	±percent r.p.m.	±percent r.p.m.			
0*	0	0.5	0.5	0.5	0.5	0.4
210*	5	.5	.5	.5	.5	.4
420*	10	.5	.5	.5	.5	.4
630	15	.5				
840*	20	.5	.5	.5	.5	.4
1,260	30	.6				
1,680*	40	.8	.6	.7	.7	.5
2,100	50	.8				
2,520*	60	.8	.6	1.0	1.0	.5
2,940	70	.8				
3,150*	75	.5	.7	1.3	1.3	.5
3,360*	80	.5	.8	1.4	1.4	.5
3,570*	85	.5	.8	1.5	1.5	.5
3,780*	90	.5	.9	1.5	1.5	.5
3,990*	95	.5	.9	1.6	1.6	.5
4,200*	100	.5	.9	1.6	1.6	.5
4,410*	105	.5	1.0	1.8	1.8	.6
4,620	110	.5				

* See 4.7.9.

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4.7.5 *Pointer alinement.* On indicators with a subdial, the indicator shall be operated so that the main pointer will indicate, respectively, 90 and 96 percent r.p.m. The subdial pointer shall correspondingly indicate 0 and 6, respectively, within 0.20 percent.

4.7.6 *Fogging.* The indicator shall be energized and placed in a 71°±2° C. controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at this temperature, an ice cube shall be rubbed on the indicator glass face for a period of 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.7.7 *Immersion (leakage).* The indicator shall be tested for leakage in accordance with Procedure I of Specification MIL–E–5272.

4.7.8 *Indicator starting at low temperature.* Prior to this test, the room temperature scale error test shall be repeated for increasing speeds only. The indicator shall then be placed in a chamber, the temperature of which is controllable. The temperature of the circulating air shall be reduced to −55°±2° C., and held at this temperature for the duration of the test. The indicator shall be kept in the chamber for at least 4 hours after the temperature of −55°±2° C. is reached before the test is started. No voltage shall be applied to the indicator terminals during this period. The indicator shall be connected to a variable frequency power supply. The voltage of the supply shall be equal to 0.004 times the frequency in cycles per minutes. The frequency of the supply shall be increased from 0 cycles per minute, and the speed at which the indicator starts to run at synchronous speed shall be noted. This speed shall not exceed 180 r.p.m. indicated speed.

4.7.9 *Scale error at −55° C.* Following the starting test at low temperature and while still at −55° C., the indicator readings at the points indicated by asterisks in tables I and II shall be determined. The frequency of the supply shall be brought up to but shall not exceed the frequency produced by a two-pole generator operating at the specified revolutions per minute. The voltage of the supply (in volts) shall be equal to approximately 0.004 times the frequency (in cycles per minute). The change in reading of the indicator from the room temperature reading at any test point shall not exceed the value specified in tables I and II.

4.7.10 *Scale error at −35° C.* The indicator readings shall be determined as specified in the scale error test at −55° C., except that the temperature of the air in the chamber shall be −35°±2° C. The indicator shall be kept at −35°±2° C. at least 2 hours prior to the test. The change in reading of the indicator from the room temperature reading at any test point shall not exceed the value specified in table I or table II.

4.7.10.1 As an alternative, this test may be repeated at temperatures of 0°±2° C. and −15°±2° C. At the test temperature of 0° C., the change in the reading of the indicator, from the room temperature reading at any test point, shall not exceed the value specified in table I or table II. At the test temperature of −15° C., the average change in reading obtained when compared with room temperature scale error, shall not exceed 0.4 percent.

4.7.11 *Indicator scale error at high temperature.* The indicator scale readings shall be determined as specified in the indicator scale error test at −55° C., except that the temperature of the air in the chamber shall be +70°±2° C. The change in reading of the indicator from the room temperature reading at any test point shall not exceed the value specified in table I or table II. The indicator shall be kept in the chamber for 4 hours after the temperature of +70° C. is reached, before the test is started.

4.7.12 *Dielectric strength.* A potential of 500 volts at commercial frequency shall be applied between isolated pins, and between pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

4.7.13 *Illumination.* The illumination of the indicator shall be tested, as required, in accordance with the detail specification.

4.7.14 *Magnetic effect.* The indicator, first not operating and then operating at an indication of approximately 90 percent r.p.m., shall

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be revolved about a short bar magnet compass with the nearest part of the indicator 5½ inches from the bar magnet, and in a vertical plane which is a perpendicular bisector of the north-south axis of the bar magnet. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 2 degrees. For integrally lighted instruments, this test shall be made with the integral lighting off and then repeated with the integral lighting on.

4.7.15 *Vibration error.* The indicator shall be tested in accordance with Procedure IV of Specification MIL–E–5272. The readings with vibration shall not vary more than 0.8 percent from the readings without vibration. The amplitude of the pointer oscillation shall not exceed 0.8 percent. The reading after vibration shall not differ from the reading before vibration by more than 0.3 percent.

4.7.16 *Indicator current.* The indicator shall be tested to determine that on a 21-volt, 4,200-c.p.m. supply, no one of the three line currents drawn by the indicator exceeds 0.30 of an ampere.

4.7.17 *Indicator temperature rise.* The indicator shall be mounted in normal dial vertical position where there is no circulation of the air surrounding the indicator, with the exception of thermal current caused by the heat dissipated from the indicator. The indicator shall be operated on a 21±0.1 volts, 4,200±5 c.p.m. supply for a period of 1 hour. The temperature rise at the end of this period, measured as described below, shall not exceed 60° C.

4.7.17.1 The temperature rise of the indicator shall be determined by change in resistance of the copper windings. The temperature rise (T–t) is calculated by the following equation:

$$T-t=\frac{(R-1)}{r}\ (234.5+t)$$

Where t = the ambient temperature and the temperature of the stator winding at the start of the test in degrees centigrade.

T = the final temperature of the stator windings, degrees centigrade.

r = the direct current resistance of the stator windings at temperature t, measured between two terminals.

R = the direct current resistance at temperature T measured between the same two terminals.

4.7.18 *Low temperature exposures.* The indicator shall be tested in accordance with Procedure II of Specification MIL–E–5272. The indicator shall then be subjected to the test specified herein for scale error at −55° C. The change in reading of the indicator from the corresponding room temperature reading shall not exceed the value specified in table I or table II. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and pass the scale error test at room temperature, increasing speeds only, and the friction test. There shall be no damage which would adversely affect subsequent operation.

4.7.19 *High temperature exposure.* The indicator shall be tested in accordance with Procedure II of Specification MIL–E–5272. It shall be subjected to and pass the individual tests specified herein. There shall be no damage which would adversely affect subsequent operation.

4.7.20 *Temperature shock.* The indicator shall be tested in accordance with Procedure I of Specification MIL–E–5272. Following this test, the indicator shall be subjected to the sealing test. No indicator leakage shall occur as a result of this test.

4.7.21 *Vibration failure.* The indicator shall be tested in accordance with Procedure V of Specification MIL–E–5272. After the 3-

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hour period, the reading for increasing speeds only shall not differ from the reading before vibration by more than 0.5 percent. In addition, no looseness in the mechanism or damage to any part of the instrument shall result from this test.

4.7.22 *Indicator overspeed and reverse operation.* One or two of the indicator mechanisms shall be electrically connected to a tachometer generator conforming to Specification MIL–G–26611 or MIL–G–9398. The reading of the indicators shall be determined when operated at 20 and 90 percent r.p.m. The indicators shall be at room temperature and the readings shall be taken before the indicators have attained an appreciable temperature rise.

4.7.22.1 The generator and indicators shall then be operated, first at a generator shaft speed of approximately 5,000 r.p.m. for a period of 5 minutes, and then in the reverse direction of rotation at a generator drive shaft speed of approximately 2,500 r.p.m. for a period of 1 minute.

4.7.22.2 After the indicators have been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 20 and 90 percent r.p.m. The change in indicated speed at the respective scale points before and after the overspeed test shall be not more than 0.5 percent.

4.7.23 *Salt spray.* The indicator shall be tested in accordance with Procedure I of Specification MIL–E–5272. The indicator shall then be subjected to the dielectric and immersion tests (4.7.12 and 4.7.7). The external surface of the indicator may be washed with distilled water and air dried prior to conducting the dielectric test.

4.7.24 *Life.* The readings of the indicator shall be determined when operated at 20 and 90 percent r.p.m. The indicator shall be at room temperature and the readings shall be taken before the indicator has attained an appreciable temperature rise. The indicator shall then be operated at an indication of 90±5 percent r.p.m. for a period of 300 hours. The voltage shall be measured at the indicator terminals and shall be 18.5±1 volts during the 300-hour period. At the end of the 300-hour period and after the indicator has been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 20 and 90 percent r.p.m. The change in reading obtained before and after this test shall not be more than 0.8 percent. No screws or other parts shall become loose or damaged as a result of this test.

4.8 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The indicator shall be packaged in accordance with level A or B of Specification MIL–P–7936, as specified in the contract or order (see 6.2). For level A packaging, preservation shall be method III of Specification MIL–P–116. Electrical connectors shall be equipped with water-vapor-proof caps and gaskets.

5.2 Packing. The indicator shall be packed in accordance with Level A, B, or C of Specification MIL–P–7936, as specified in the contract or order (see 6.2).

5.3 Marking. The interior and exterior containers shall be marked as specified in Specification MIL–P–7936.

5.4 Reinspection dates. Reinspection markings shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indicating the turbine speed, in percent, of turbine engines. The indicator calibration is based on the tachometer generator turning at 4,200 r.p.m. at 100 percent of rated turbine speed.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of the detail specification (see 3.1).
(b) Whether sampling plan B is to be omitted (see 4.4.2.2).
(c) Name of the laboratory where samples should be submitted (see 4.3.1).
(d) Levels of packaging and packing desired (see 5.1 and 5.2).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division (ASD), Attn: ASNPFC, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

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